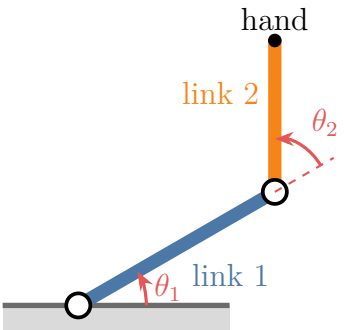
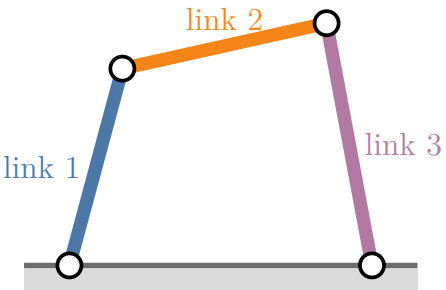
2R arm on the robot



ground: the robot's top plate (link 0)

$N = 3$ links (with ground)
$J = 2$ revolute joints
$3\,(3 - 1 - 2) + 2 = 2$

a closed loop of links



ground (link 0)

$N = 4$ links (with ground)
$J = 4$ revolute joints
$3\,(4 - 1 - 4) + 4 = 1$

open circle: a revolute joint, which lets one link turn about the other (1 freedom); flat motion: $m = 3$